

Multiple Choice (10 marks)

1. Some recent work has found an association between obesity and
 - (a) Shorter telomere strands
 - (b) Tobacco use
 - (c) Excess alcohol use
 - (d) Skin cancers
2. The DiGeorge/Shprintzen syndrome is caused by a deletion in which chromosome?
 - (a) 4
 - (b) 7
 - (c) 15
 - (d) 22
3. NEAT may be one way in which some people are able to maintain a healthy BMI; NEAT refers to
 - (a) Eating the right kinds of low calorie foods
 - (b) Daily activities like walking, gesturing, and even fidgeting
 - (c) Intense aerobic exercise practiced at least 3 times a week
 - (d) Surgical removal of fatty tissue
4. The Apo-E₂ protein is very prevalent in people who
 - (a) Die young
 - (b) Live to be 100 or older
 - (c) Have healthy immune systems
 - (d) Also have high levels of Apo-E₄
5. What is the genetic structure of rotavirus?
 - (a) Circular DNA containing virus
 - (b) RNA virus with RT
 - (c) ds RNA virus
 - (d) ds RNA virus in segments

True / False (10 marks)

Answer True or False

6. True or False: The chemistry of the DNA and RNA is the most important factor for virus classification.
7. True or False: Orders of monks and nuns in various religions practice asceticism, emphasizing celibacy, virginity, and poverty.
8. True or False: The RT PCR kit is essential for the diagnosis of arenavirus infection.
9. True or False: Condom use has no effect on the risk of acquiring an HIV infection from an infected partner.
10. True or False: Cerebrospinal fluid circulates between the arachnoid and pia maters surrounding the brain.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the embryological development of the hyoid bone, detailing the contributions of the second and third pharyngeal arches and their significance in human anatomy.
12. Discuss the appropriate study design an investigator should use to explore the potential link between first trimester acetaminophen use and neural tube defects, considering the estimated risk in the general population.

End of Paper